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Computation Offloading Toward Edge Computing

A survey of recent research efforts made towards offloading in edge computing.
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Workshop

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Introduction

Addressing the growing demands of mobile computing with edge intelligence.

Growing Demands of Mobile

Mobile devices like smartphones, wearables, and IoT are becoming more popular, but they have limited computational power and battery life

Emerging applications such as Augmented Reality (AR), Artificial Intelligence, and Cloud gaming demand more resources

Offloading computation to cloud resources helps overcome these limitations. One big issue: **Latency**

A shift towards Edge

Edge computing keeps processing and analysis near the edge of the network

Key benefits: Ultra-low latency, real-time decision-making, and reduced bandwidth usage.

Suitable for real-time applications such as Autonomous driving which require very low latency, which would not be possible with public cloud(> 100ms)

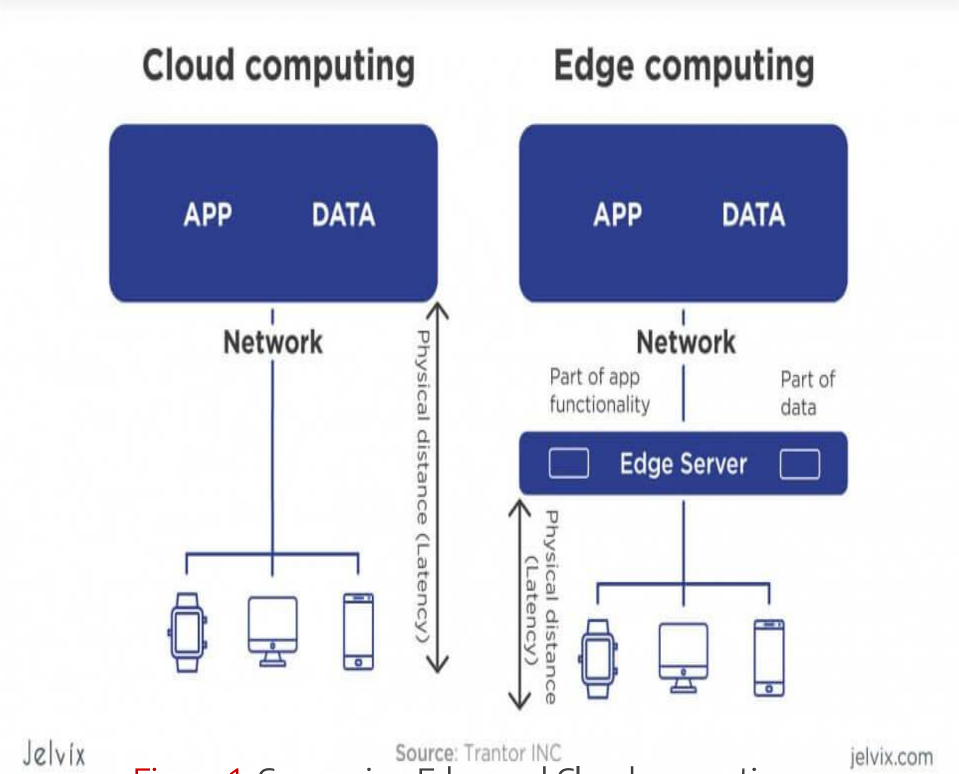

Figure 1: Comparing Edge and Cloud computing

Image Source: Jelvix, https://jelvix.com/blog/what-is-edge-computing

Time to rethink offloading?

Task placement: Mobile, Edge or the Cloud?

Task can now be offloaded to either edge or the cloud, making offloading decisions more complicated

Task execution in a heterogenous setting?

Execution of tasks is now distributed among multiple edge nodes and the cloud, requiring developers to rethink code partitioning

Deployment standardization?

Unlike data centres, there is no standardized deployment for edge nodes

A discussion on these issues is required

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Background and Motivation

From Cloud to the Edge: a paradigm shift

Evolution of computational offloading

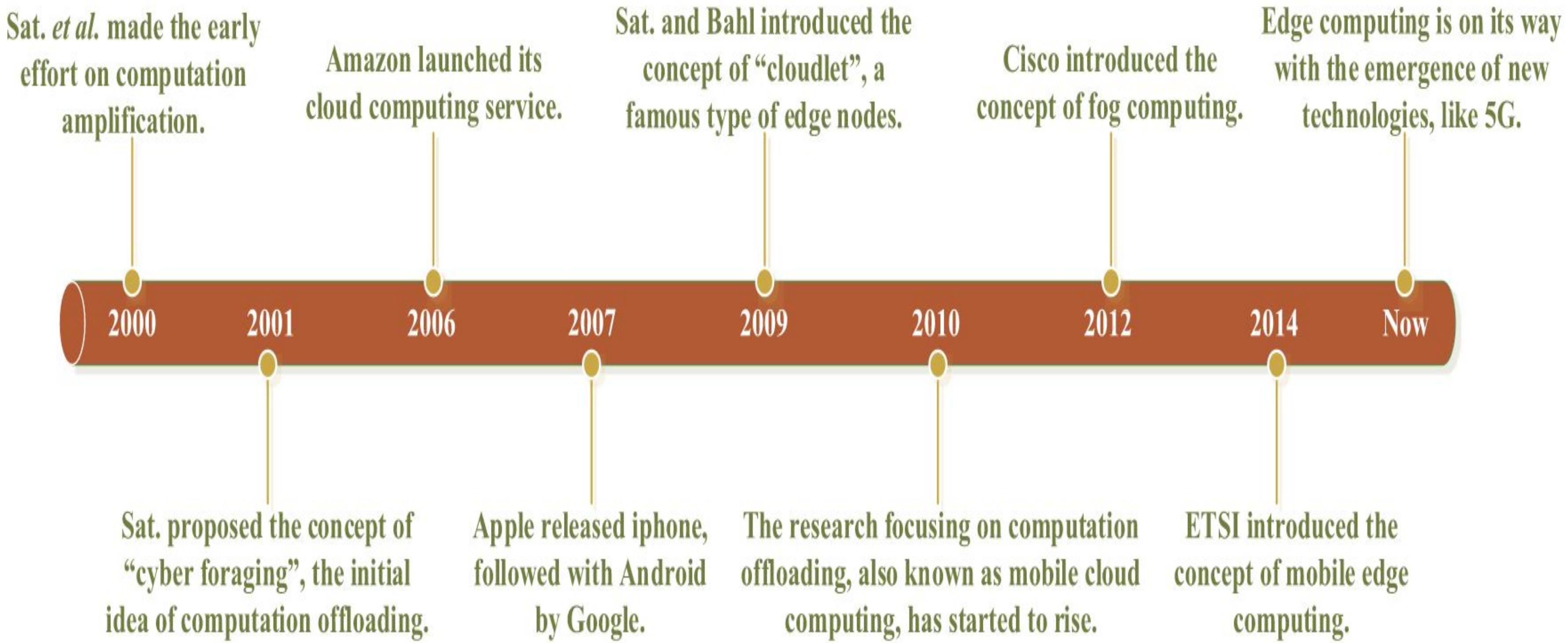

Figure 2: Evolution of computational offloading

From cloud to Edge Computing

Cloud Computing (2000s)

- Centralized model offering large-scale computation
- High latency due to distance

Mobile Cloud Computing (2010s)

- Mobile devices offload task to cloud
- Still faced latency and bandwidth challenges

Edge Computing (Present)

- Decentralized, proximity-based model
- Reduces latency and core traffic

Need for Edge Computing

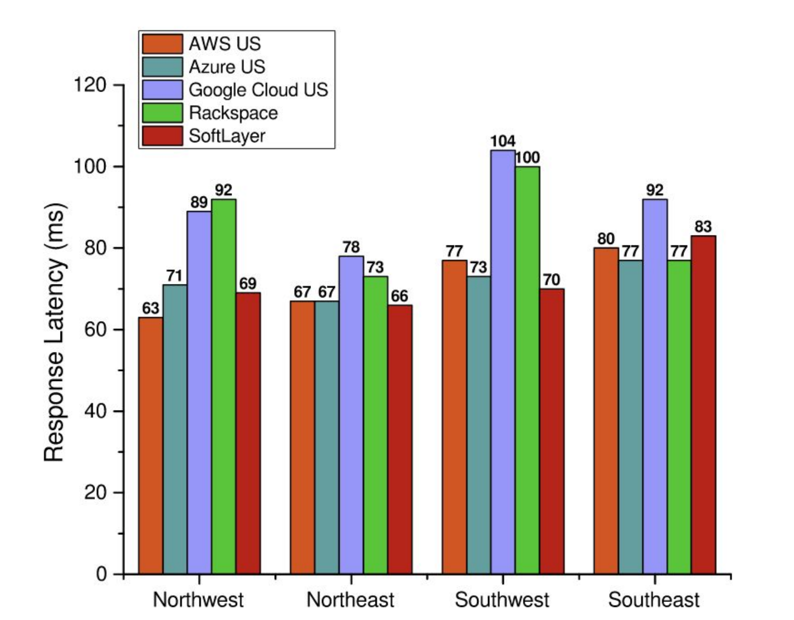

Figure 3: Average response latency of five top public cloud providers in the United States for serving in four regions of the country. Source from Cedexis/Network World

Latency

Emerging applications are latency-sensitive, public clouds have high latency

Improved Data Rates

5G offers high data capacity that can allow processing within Radio Access Networks (RAN)

Reduce core traffic

Computing at edge can reduce traffic on core infrastructures

Related Concepts

- Computational Offloading
- Edge Computing
- Fog Computing
- 5G

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Overview of Edge Computing

Architecture, Data flow, and Granularity

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A Three-tier Architecture

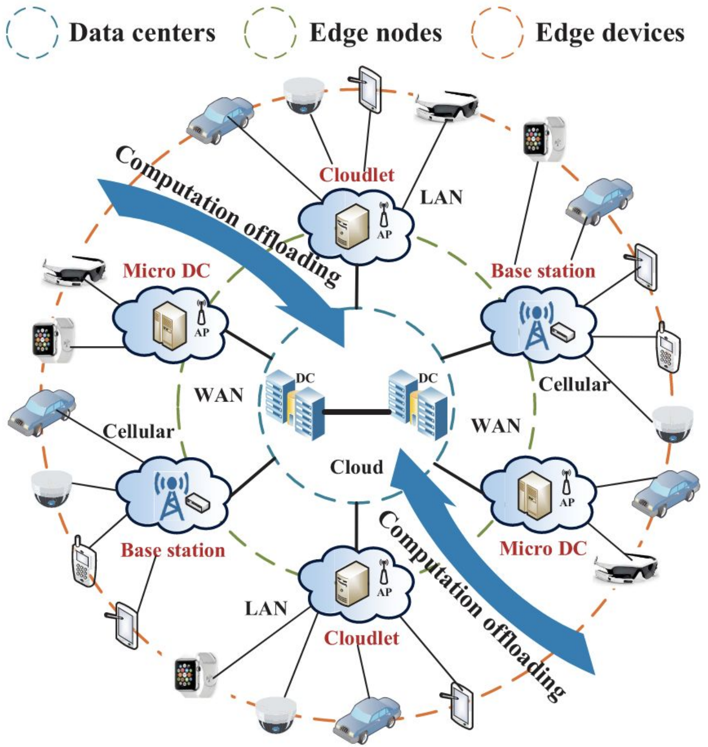


Figure 4: Architecture of Edge Computing with flow of computation

Components

Edge Devices

Smartphones, vehicles, IoT devices that generate data

Edge nodes

Middle layer that provides a link between edge devices and the core

Data Centres

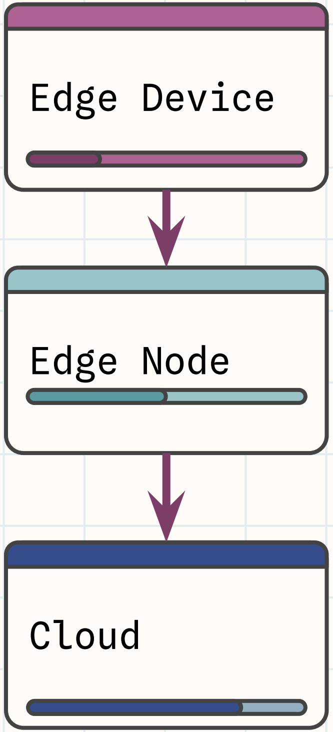
Core of the network providing resource discovery and management, as well as global big data analytics

Role of Edge Nodes

Node

- **Cloudlets**
  - Small resource rich boxes to provide proximity computing
  - Deployed in public spaces
- **Base Stations**
  - Provide computing service in addition to communication functions
  - Eg: 5G base stations
- **Micro Data Centers**
  - Geo-distributed small Data centers small or moderate order of servers
  - Provide localized computation
- **Other types**
  - Any infrastructure connecting edge devices to cloud

Data Flow



Data generated on device (e.g., video capture).

Tasks offloaded for faster processing (e.g., object detection).

Results or metadata for large-scale analysis or storage. (e.g., event timestamp)

Analysis of Offloading:

Computation offloading happens only if the time of the local execution is longer than its total offloading time

$Tlocal > Toffloading = Tcomm + Tremote$

Also considering Energy, the energy consumption of data transmission is smaller than local execution

$Elocal > Ecomm$

Granularity of Offloading

(a)Full (b)Task (c)Method/Thread

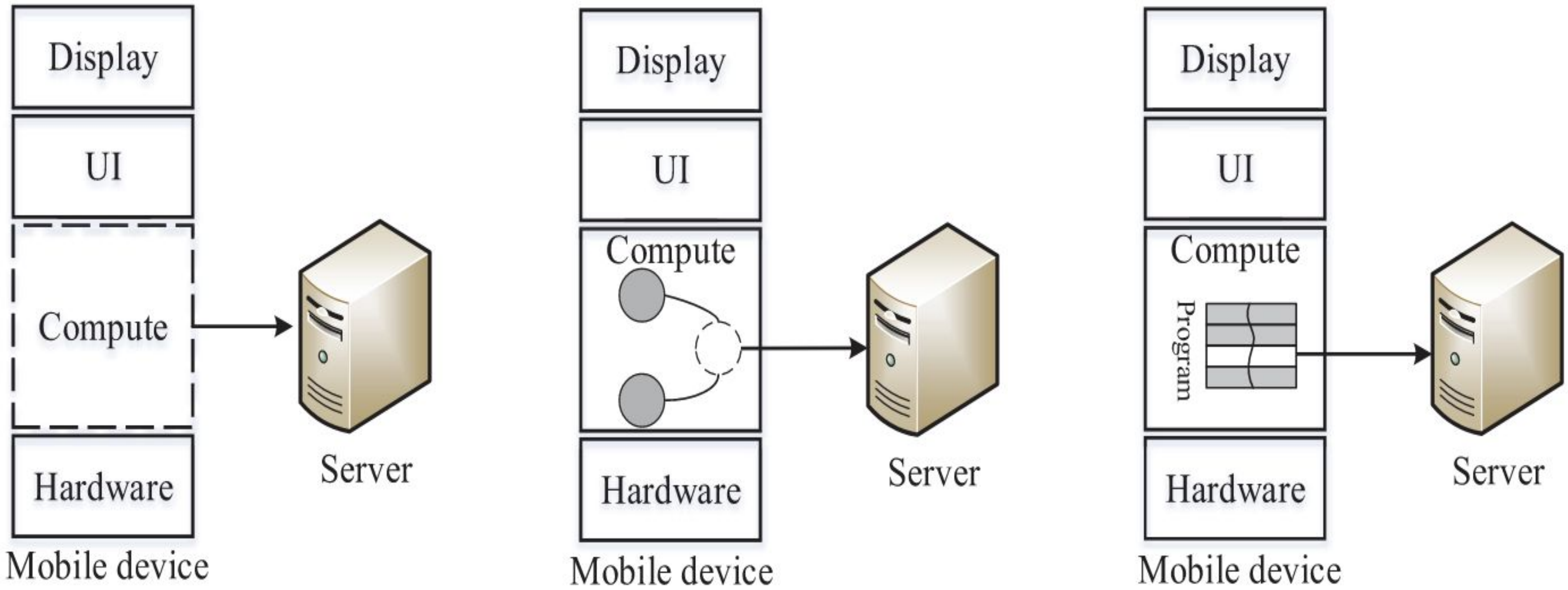


Figure 5: Granularity of computation offloading

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Challenges

An investigation into challenges emerging in Edge

Complexities in Computation Offloading

Why offloading is challenging?

- Diverse and dynamic environments (heterogeneous devices, networks and nodes)
- Real-time requirements for latency sensitive tasks

Problems

Partitioning
Deciding what to offload

Task Allocation
Where and how to offload tasks

Task Execution
Managing execution across heterogeneous platforms

Application Partitioning

Divide the application code into several parts that will be executed on different platforms

Mobile-cloud

- Mobile cloud computing (MCC), introduced partitioning as either **automatic** through program analysis or through **programmer-specified** partitioning

Edge Computing

- In Edge computing, we have to consider new challenges; **multinode offloading** and **partitioning granularity**

Application Partitioning: Edge

Divide the application code into several parts that will be executed on different platforms

Multinode Offloading

- In MCC, tasks have a specific site to be placed (mobile or cloud)
- What do you do when your application has to be split across multiple edge nodes, or even clouds?
- A partitioning algorithm is required for task distribution

Granularity

- Method, thread and class level partitioning in MCC requires data and context dependence
- Expensive in heterogeneous edge environments
- Offloading at **task/component level** provides better efficiency
- Eg: Face Detection

Task Allocation

Runtime decision of task placement and scheduling associated with resource management

Cloudlet Discovery

- Cloudlets are distributed and not well-organized
- Identify nearby cloudlets to minized user response times
- Discovery mechanisms: LAN Discovery and WAN Discovery (Ha)
- **Application-aware** (eg: gaming vs vision) and **Service-aware** discovery
- **Cloudlet placement**: problems of placing cloudlets optimally

Task Scheduling with Resource Management

- **Goal**: Minimize energy, balance loads, and improve QoS in dynamic environments.
- **Challenges:** Multiuser and multinode coordination, dynamic resource allocation, rapid changes from user mobility, and limited cloudlet capacities.

Task Execution

Executing Tasks Across Distributed Platforms

Challenges

- **Heterogeneous Platforms**:
  - Mobile devices (ARM-based) differ from Edge Nodes (x86)
- **Mobility and continuity**:
  - Devices may move between edge nodes
  - Requires seamless task migration and handover
- **Latency-sensitive tasks**

Solutions and trends

VM-based execution

Offers isolation but is resource intensive.

Containers

Lightweight alternatives for faster deployment and reduced resource usage.

Serverless

On-demand task execution tailored for dynamic workloads.

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Applications and Use Cases

Some typical application scenarios that benefit from edge computing

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Cloud Gaming

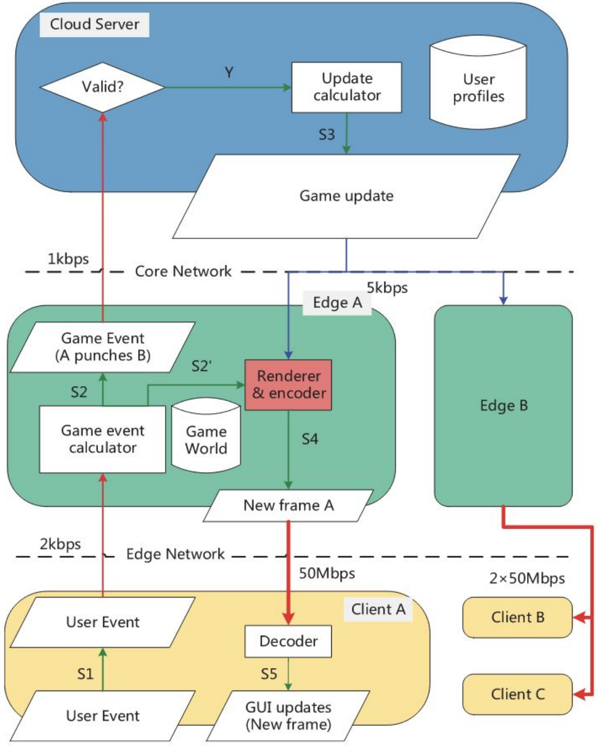

Figure 6: Workflow of EC+

- What is cloud gaming?
- Mobile devices offload game running and rendering to the cloud.
- Removes platform dependency and mobile device limitations

- Types:
  - **Video Streaming** (Full offloading)
  - **Graphics Streaming**(Partial Offloading)

- Challenges:
  - Network latency for FPS games
  - High Bandwidth demands

- Solution: Hybrid edge-cloud architectures (e.g., **EC+**), offload latency sensitive task to edge, reducing latency and network traffic.

Video analytics

Figure 7: Video analytics using cameras

- Real-Time video Analytics is now possible with edge computing due to the explosion of cameras. Eg: Autonomous vehicles, AR
- Challenges:
  - Compute-intensive and bandwidth hungry tasks
  - Limited device capabilities and high latency in cloud-only solutions
- Edge Computing Solutions:
  - Three-tier architecture
    - Lightweight processing at edge devices
    - Moderate processing at edge nodes
    - Big Data processing in the cloud

Video analytics

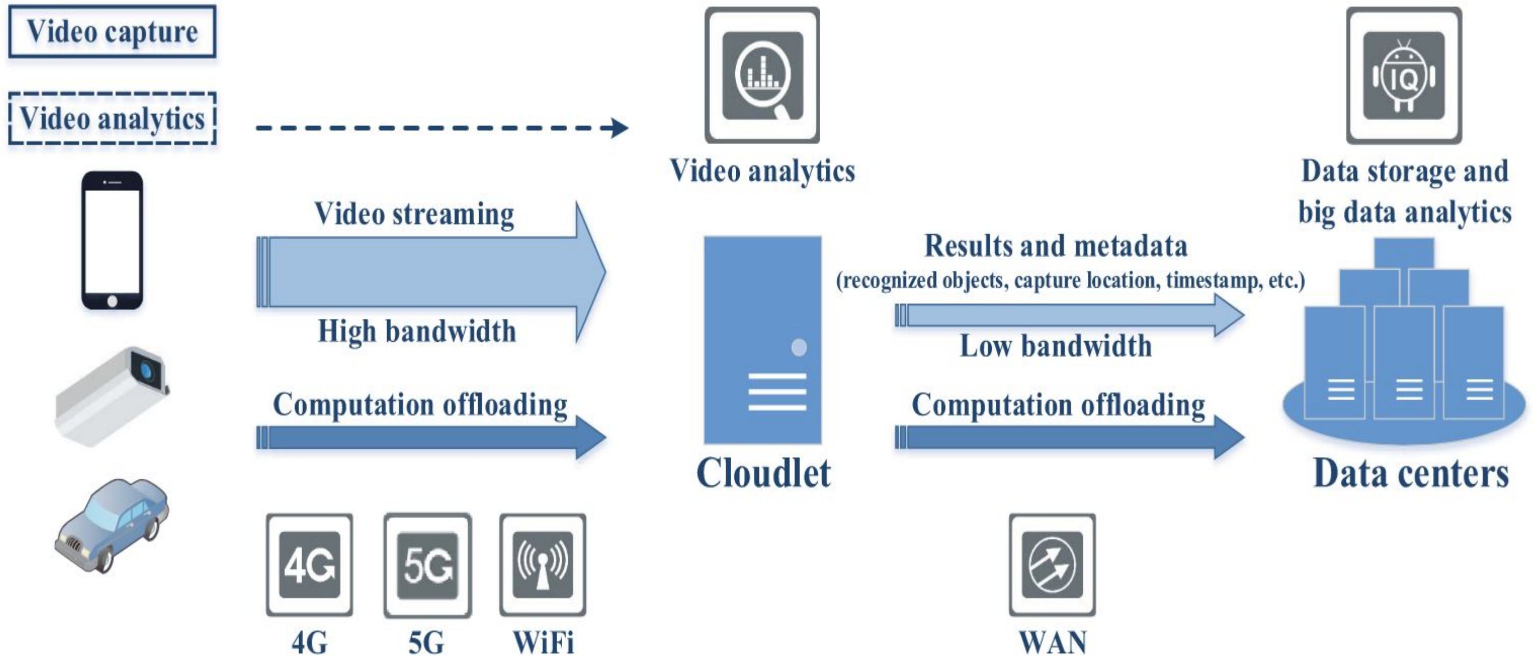

Figure 8: Three-tier architecture for video analytics

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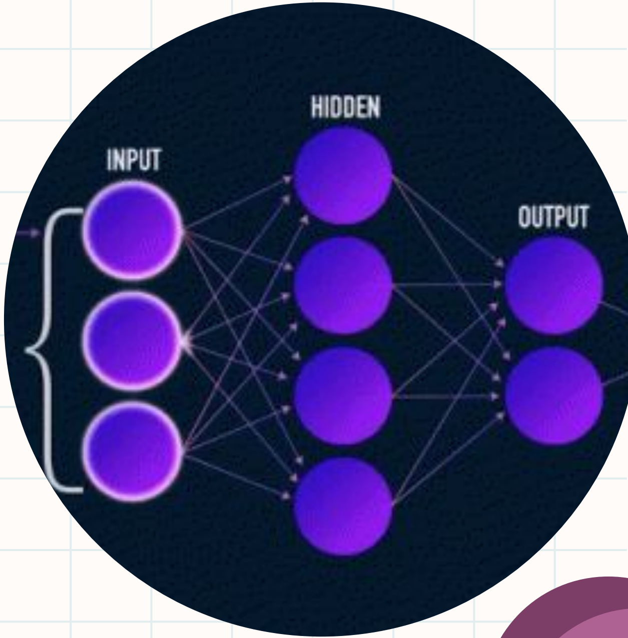
Future Directions

Trends moving forward

Proposed Innovations

- Using AI powered techniques to optimize edge-cloud amalgamation:
  - AI-Powered Data Prioritization and Routing
  - Distributed Edge Analytics and Machine Learning
  - Predictive Resource Allocation
  - Automated Cloud-Edge Orchestration
  - Domain-Specific Data Processing Pipelines
  - Intelligent Offloading Framework
- Offloading-as-a-service
- Security and Privacy



Intelligent Offloading Framework

"To develop an intelligent cloud offloading framework that optimizes task distribution between edge and cloud environments based on real-time conditions to maximize computational efficiency and minimize latency and resource consumption."

Challenges to Address:

- **Dynamic Conditions**: Adapting to changes in latency, bandwidth, and resource availability.
- **Task Partitioning**: Deciding how much of a task to offload.
- **Heterogeneity**: Managing diverse platforms across edge and cloud.

Future Directions

- Edge computing is a distributed paradigm where computation occurs close to data sources
- Reduces latency and distributes network traffic
- **Key Challenges**:
  - Application Partitioning
  - Task allocation and execution
  - Resource management
- **Potential Technologies**:
  - Containerization and Serverless
  - Artificial intelligence
  - 5G/6G

- A seamless flow of computation across edge devices, edge nodes, and cloud infrastructure.

Thanks!

Do you have any questions?